Serii de puteri

• Să se determine raza de convergență și mulțimea de convergență:

1. $\sum_{n\geq 1}(-1)^{n+1}\frac{x^n}{n}=\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}x^n$. Notăm: $a_n=\frac{(-1)^{n+1}}{n}$

$$L=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(-1)^{n+2}}{n+1}\cdot\frac{n}{(-1)^{n+1}}\right|=\lim_{n\to\infty}\left|\frac{-n}{n+1}\right|=$$

$$=\lim_{n\to\infty}\frac{n}{n+1}=1\Rightarrow R=\frac{1}{L}=1$$

pt $x=1$: $\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}=\sum_{n\geq 1}(-1)^n\cdot\frac{-1}{n}$. Notăm: $b_n=\frac{-1}{n}$

și folosim criteriul Leibnitz:

$$\frac{b_{n+1}}{b_n}=-\frac{1}{n+1}\cdot\frac{n}{-1}=\frac{n}{n+1}<1\Rightarrow b_{n+1}<b_n\Rightarrow b_n\downarrow\ (1)$$

$$\lim_{n\to\infty}b_n=\lim_{n\to\infty}\frac{-1}{n}=0\ (2)$$

$\overset{(1)(2)}{\Longrightarrow}\sum_{n\geq 1}(-1)^n\cdot\frac{-1}{n}$ convergentă.

pt $x=-1$: $\sum_{n\geq 1}\frac{(-1)^{n+1}}{n}\cdot(-1)^n=\sum_{n\geq 1}\frac{(-1)^{2n+1}}{n}=\sum_{n\geq 1}\frac{-1}{n}=$

$=-\sum_{n\geq 1}\frac{1}{n}$ divergentă (serie armonică generalizată cu $\alpha=1$)

$\Rightarrow$ M.C.: $(-1,1]$

2. $\sum_{n\geq 1}\frac{n^n x^n}{n!}=\sum_{n\geq 1}\frac{n^n}{n!}x^n$. Notăm: $a_n=\frac{n^n}{n!}$

$$L=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(n+1)^{n+1}}{(n+1)!}\cdot\frac{n!}{n^n}\right|=\lim_{n\to\infty}\left(\frac{n+1}{n}\right)^n=$$

$$=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n=e\Rightarrow R=\frac{1}{L}=\frac{1}{e}$$

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pt $x = \frac{1}{e}$: $\sum_{n\geq 1} \frac{n^n \left(\frac{1}{e}\right)^n}{n!}$ Notăm: $x_n = \frac{n^n \left(\frac{1}{e}\right)^n}{n!}$.

Folosim criteriul Raabe - Duhamel:

$$\lim_{n\to\infty} n\left(\frac{x_n}{x_{n+1}} - 1\right) = \lim_{n\to\infty} n\left(\frac{n^n \left(\frac{1}{e}\right)^n}{n!} \cdot \frac{(n+1)!}{(n+1)^{n+1}\left(\frac{1}{e}\right)^{n+1}} - 1\right) =$$

$$= \lim_{n\to\infty} n\left[\underbrace{\left(\frac{n}{n+1}\right)^n}_{\to 1/e} \cdot \frac{1}{e} - 1\right] = -\infty < 1 \Rightarrow$$

$\Rightarrow \sum_{n\geq 1} x_n$ divergentă.

pt $x = -\frac{1}{e}$: $\sum_{n\geq 1} \frac{n^n \left(-\frac{1}{e}\right)^n}{n!}$. Notăm $y_n = \frac{n^n \left(-\frac{1}{e}\right)^n}{n!}$

caz I: $n = 2k \Rightarrow y_{2k} = \frac{(2k)^{2k}\left(-\frac{1}{e}\right)^{2k}}{(2k)!} = \frac{(2k)^{2k}\left(\frac{1}{e}\right)^{2k}}{(2k)!} =$

$= x_n$

$\Rightarrow \sum_{n\geq 1} y_{2k}$ divergentă

caz II: $n = 2k+1 \Rightarrow y_{2k+1} = \frac{(2k+1)^{2k+1}\left(-\frac{1}{e}\right)^{2k+1}}{(2k+1)!}$

..... (Raabe - Duhamel $\Rightarrow$ divergentă)

$\Rightarrow$ M.C.: $\left(-\frac{1}{e}, \frac{1}{e}\right)$

3. $\sum_{n\geq 0} \left[2 + (-1)^n\right] x^n$. Notăm: $a_n = 2 + (-1)^n$

$$l = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{|2 + (-1)^n|} = 1 \Rightarrow R = \frac{1}{l} = 1.$$

pt $x = 1$: $\sum_{n\geq 0} 2 + (-1)^n$. Notăm: $x_n = 2 + (-1)^n$ și

Folosim criteriul necesar:

$n = 2k \Rightarrow x_n \xrightarrow{n\to\infty} 3$

$n = 2k+1 \Rightarrow x_n \xrightarrow{n\to\infty} 1$ $\Rightarrow \nexists \lim_{n\to\infty} x_n \Rightarrow \sum_{n\geq 0} x_n$ divergentă

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pt $x=-1$: $\sum_{n\geq 0}\left[2+(-1)^n\right](-1)^n = \sum_{n\geq 0}\left[2(-1)^n+1\right]$

Notăm: $y_n = 2(-1)^n+1$ și folosim criteriul necesar:

$n=2k \Rightarrow y_n \xrightarrow{n\to\infty} 3$

$n=2k+1 \Rightarrow y_n \xrightarrow{n\to\infty} -1$

$\Rightarrow \nexists \lim_{n\to\infty} y_n \Rightarrow \sum_{n\geq 0} y_n$ divergentă

$\Rightarrow$ M.C.: $(-1;1)$

4. $\sum_{n\geq 0} \frac{n+1}{\sqrt{n^4+n^3+1}}\left(\frac{x+1}{2x+3}\right)^n$; $x=-\frac{3}{2}$

Notăm: $y = \frac{x+1}{2x+3} \Rightarrow \sum_{n\geq 0}\frac{n+1}{\sqrt{n^4+n^3+1}}y^n$.

Notăm: $\frac{n+1}{\sqrt{n^4+n^3+1}} = a_n$

$$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{n+2}{\sqrt{(n+1)^4+(n+1)^3+1}}\cdot\frac{\sqrt{n^4+n^3+1}}{n+1}\right| =$$

$$= \lim_{n\to\infty}\underbrace{\frac{n+2}{n+1}}_{\to 1}\cdot\underbrace{\frac{\sqrt{n^4+n^3+1}}{\sqrt{(n+1)^4+(n+1)^3+1}}}_{\to 1} = 1 \Rightarrow R = \frac{1}{L} = 1$$

pt $y=1$: $\sum_{n\geq 0}\frac{n+1}{\sqrt{n^4+n^3+1}}$. Notăm: $b_n = \frac{n+1}{\sqrt{n^4+n^3+1}}$ și folosim criteriul comparației la limită.

Notăm: $u_n = \frac{1}{n^\alpha}$; $\alpha\in\mathbb{R}$.

$$\Rightarrow \lim_{n\to\infty}\frac{b_n}{u_n} = \lim_{n\to\infty}\frac{n^{\alpha+1}\left(1+\frac{1}{n}\right)}{n^2\sqrt{1+\frac{1}{n}+\frac{1}{n^4}}} = \lim_{n\to\infty} n^{\alpha-1}\frac{1+\underbrace{\frac{1}{n}}_{\to 0}}{\sqrt{1+\underbrace{\frac{1}{n}}_{\to 0}+\underbrace{\frac{1}{n^4}}_{\to 0}}} \neq$$

$\neq 0$ și $\neq +\infty \Leftrightarrow \alpha = 1 \Rightarrow$

$\Rightarrow \sum_{n\geq 0} u_n$ divergentă $\Rightarrow \sum_{n\geq 0} b_n$ divergentă.

pt $y=-1$: $\sum_{n\geq 0}\frac{n+1}{\sqrt{n^4+n^3+1}}(-1)^n$. Notăm $c_n = \frac{n+1}{\sqrt{n^4+n^3+1}}$ și folosim criteriul Leibnitz:

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Indexăm funcția $f:[1,\infty)\to\mathbb{R}$; $f(x)=\frac{x+1}{\sqrt{x^4+x^3+1}}$.

$$f'(x)=\frac{\sqrt{x^4+x^3+1}-\frac{(x+1)(4x^3+3x^2)}{2\sqrt{x^4+x^3+1}}}{x^4+x^3+1}=$$

$$=\frac{2(x^4+x^3+1)-4x^4-4x^3-3x^3-3x^2}{2(x^4+x^3+1)\sqrt{x^4+x^3+1}}=\frac{-2x^4-5x^3-3x^2+2}{2(x^4+x^3+1)\sqrt{x^4+x^3+1}}<1;\ \forall x\in[1,\infty)$$

$\Rightarrow f\searrow$ pe $[1,\infty)\Rightarrow 1\le m<m+1\Leftrightarrow f(m)>f(m+1)\Leftrightarrow$

$\Leftrightarrow c_m>c_{m+1}\Rightarrow c_m\searrow$ (1)

$$\lim_{n\to\infty}c_n=\lim_{n\to\infty}\frac{n+1}{\sqrt{n^4+n^3+1}}=\lim_{n\to\infty}\frac{n\left(1+\frac{1}{n}\right)}{n^2\sqrt{1+\frac{1}{n}+\frac{1}{n^4}}}=0\ (2)$$

$\overset{(1)(2)}{\Longrightarrow}$ $\sum_{n\ge1}(-1)^n\frac{n+1}{\sqrt{n^4+n^3+1}}$ convergentă

$-1\le y<1\Leftrightarrow -1\le\frac{x+1}{2x+3}<1$

I $-1\le\frac{x+1}{2x+3}\Leftrightarrow\frac{x+1}{2x+3}+1\ge0\Leftrightarrow\frac{3x+4}{2x+3}\ge0\Leftrightarrow$

$$\Leftrightarrow\begin{cases}\begin{cases}3x+4\ge0\Rightarrow x\ge-\frac{4}{3}\Rightarrow x\in\left[-\frac{4}{3},\infty\right)\\ 2x+3>0\Rightarrow x>-\frac{3}{2}\Rightarrow x\in\left(-\frac{3}{2},\infty\right)\end{cases}\overset{\cap}{\Rightarrow}x\in\left[-\frac{4}{3},\infty\right)\\ \begin{cases}3x+4\le0\Rightarrow x\le-\frac{4}{3}\Rightarrow x\in\left(-\infty,-\frac{4}{3}\right]\\ 2x+3<0\Rightarrow x<-\frac{3}{2}\Rightarrow x\in\left(-\infty,-\frac{3}{2}\right)\end{cases}\overset{\cap}{\Rightarrow}x\in\left(-\infty,-\frac{3}{2}\right)\end{cases}\overset{\cup}{\Longrightarrow}$$

$\Rightarrow x\in\left(-\infty,-\frac{3}{2}\right)\cup\left(-\frac{4}{3},\infty\right)$ (1)

II $\frac{x+1}{2x+3}<1\Leftrightarrow\frac{x+1}{2x+3}-1<0\Leftrightarrow\frac{-x-2}{2x+3}<0\Leftrightarrow$

$$\Leftrightarrow\begin{cases}\begin{cases}-x-2<0\Rightarrow x>-2\Rightarrow x\in(-2,\infty)\\ 2x+3>0\Rightarrow x>-\frac{3}{2}\Rightarrow x\in\left(-\frac{3}{2},\infty\right)\end{cases}\overset{\cap}{\Rightarrow}x\in\left(-\frac{3}{2},\infty\right)\\ \begin{cases}-x-2>0\Rightarrow x<-2\Rightarrow x\in(-\infty,-2)\\ 2x+3<0\Rightarrow x<-\frac{3}{2}\Rightarrow x\in\left(-\infty,-\frac{3}{2}\right)\end{cases}\overset{\cap}{\Rightarrow}x\in(-\infty,-2)\end{cases}\overset{\cup}{\Longrightarrow}$$

$\Rightarrow x\in(-\infty,-2)\cup\left(-\frac{3}{2},\infty\right)$ (2)

$\overset{(1)(2)}{\underset{\cap}{\Longrightarrow}}$ M.C.: $x\in(-\infty,-2)\cup\left(-\frac{4}{3},\infty\right)$

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5. $\sum_{n\geq 0}(-1)^n \frac{\sqrt{n^2+1}}{n^2+n+1}\left(\frac{4x-1}{x+3}\right)^n$. Notăm: $y=\frac{4x-1}{x+3}$

$\Rightarrow \sum_{n\geq 0}(-1)^n \frac{\sqrt{n^2+1}}{n^2+n+1} y^n$. Notăm: $a_n=(-1)^n\frac{\sqrt{n^2+1}}{n^2+n+1}$

$L=\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(-1)^{n+1}\sqrt{(n+1)^2+1}}{(n+1)^2+n+1+1}\cdot\frac{n^2+n+1}{(-1)^n\sqrt{n^2+1}}\right|=$

$=\lim_{n\to\infty}\frac{(n^2+n+1)\sqrt{(n+1)^2+1}}{\left[(n+1)^2+n+2\right]\sqrt{n^2+1}}=\lim_{n\to\infty}\frac{n^2\left(1+\frac{1}{n}+\frac{1}{n^2}\right)\sqrt{\left(1+\frac{1}{n}\right)^2+\frac{1}{n^2}}}{n^2\left[\left(1+\frac{1}{n}\right)^2+\frac{1}{n}+\frac{2}{n^2}\right]\sqrt{1+\frac{1}{n^2}}}=1$

$\Rightarrow R=\frac{1}{L}=1$

pt $y=1$: $\sum_{n\geq 0}(-1)^n\frac{\sqrt{n^2+1}}{n^2+n+1}$. Notăm: $b_n=\frac{\sqrt{n^2+1}}{n^2+n+1}$

Indexăm funcția $f:[0,\infty)\to\mathbb{R}$; $f(x)=\frac{\sqrt{x^2+1}}{x^2+x+1}$

$f'(x)=\frac{\frac{x(x^2+x+1)}{\sqrt{x^2+1}}-\sqrt{x^2+1}(2x+1)}{(x^2+x+1)^2}=\frac{x^3+x^2+x-(x^2+1)(2x+1)}{(x^2+x+1)^2\sqrt{x^2+1}}=$

$=\frac{x^3+x^2+x-2x^3-2x-x^2-1}{(x^2+x+1)^2\sqrt{x^2+1}}=\frac{-x^3-x-1}{(x^2+x+1)^2\sqrt{x^2+1}}<0;\ \forall x\in[0,\infty)$

$\Rightarrow f$ ↓s pe $[0,\infty)\Rightarrow 0\leq n<n+1\Leftrightarrow f(n)>f(n+1)\Leftrightarrow$

$\Leftrightarrow b_n>b_{n+1}\Leftrightarrow b_n$ ↓s (1)

$\lim_{n\to\infty} b_n=\lim_{n\to\infty}\frac{\sqrt{n^2+1}}{n^2+n+1}=\lim_{n\to\infty}\frac{n\sqrt{1+\frac{1}{n^2}}}{n^2\left(1+\frac{1}{n}+\frac{1}{n^2}\right)}=0$ (2)

$\xrightarrow[\text{Leibnitz}]{(1)(2)}\sum_{n\geq 0}(-1)^n b_n$ convergentă

pt $y=-1$: $\sum_{n\geq 0}\frac{\sqrt{n^2+1}}{n^2+n+1}$. Folosim criteriul comparației la limită. Fie $c_n=\frac{1}{n^\alpha}$; $\alpha\in\mathbb{R}$

$\Rightarrow \lim_{n\to\infty}\frac{b_n}{c_n}=\lim_{n\to\infty}\frac{n^\alpha\sqrt{n^2+1}}{n^2+n+1}=\lim_{n\to\infty}\frac{n^{\alpha+1}\sqrt{1+\frac{1}{n}}}{n^2\left(1+\frac{1}{n}+\frac{1}{n^2}\right)}=$

$=\lim_{n\to\infty} n^{\alpha-1}\cdot\underbrace{\frac{\sqrt{1+\frac{1}{n}}}{1+\frac{1}{n}+\frac{1}{n^2}}}_{\to 1}\neq 0$ și $\neq\infty\Leftrightarrow\alpha=1\Rightarrow$

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$\Rightarrow \sum_{n\geq 0} c_n$ divergentă $\Rightarrow \sum_{n\geq 0} b_n$ divergentă

$\Rightarrow -1 < y \leq 1 \Leftrightarrow -1 < \frac{4x-1}{x+3} \leq 1$

$\text{I}: -1 < \frac{4x-1}{x+3} \Leftrightarrow \frac{4x-1}{x+3} + 1 > 0 \Leftrightarrow \frac{5x+2}{x+3} > 0 \Leftrightarrow$

$\Leftrightarrow \begin{cases} \begin{cases} 5x+2>0 \Rightarrow x > -\frac{2}{5} \Rightarrow x \in (-\frac{2}{5}, \infty) \\ x+3>0 \Rightarrow x > -3 \Rightarrow x \in (-3, \infty) \end{cases} \Rightarrow x \in (-\frac{2}{5}, \infty) \\ \begin{cases} 5x+2<0 \Rightarrow x < -\frac{2}{5} \Rightarrow x \in (-\infty, -\frac{2}{5}) \\ x+3<0 \Rightarrow x < -3 \Rightarrow x \in (-\infty, -3) \end{cases} \Rightarrow x \in (-\infty, -3) \end{cases} \overset{\cup}{\Rightarrow}$

$\Rightarrow x \in (-\infty, -3) \cup (-\frac{2}{5}, \infty)$ (1)

$\text{II}: \frac{4x-1}{x+3} \leq 1 \Leftrightarrow \frac{4x-1}{x+3} - 1 \leq 0 \Leftrightarrow \frac{3x-4}{x+3} \leq 0 \Leftrightarrow$

$\Leftrightarrow \begin{cases} \begin{cases} 3x-4 \leq 0 \Rightarrow x \leq +\frac{4}{3} \Rightarrow x \in (-\infty, +\frac{4}{3}] \\ x+3>0 \Rightarrow x > -3 \Rightarrow x \in (-3, \infty) \end{cases} \Rightarrow x \in (-3, +\frac{4}{3}] \\ \begin{cases} 3x-4 \geq 0 \Rightarrow x \geq +\frac{4}{3} \Rightarrow x \in [+\frac{4}{3}, \infty) \\ x+3<0 \Rightarrow x < -3 \Rightarrow x \in (-\infty, -3) \end{cases} \Rightarrow x \in \emptyset \end{cases} \overset{\cup}{\Rightarrow}$

$\Rightarrow x \in (-3, +\frac{4}{3}]$ (2)

$\overset{(1)(2)}{\underset{\cap}{\Longrightarrow}}$ M.C.: $x \in (-\frac{2}{5}, \frac{4}{3}]$

- Să se determine mulțimea de convergență și suma seriilor:

1. $\sum_{n\geq 0} (-1)^n \cdot \frac{x^{2n+1}}{2n+1}$. Notăm: $a_n = \frac{(-1)^n}{2n+1}$

$L = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \frac{(-1)^{n+1}}{2n+3} \cdot \frac{2n+1}{(-1)^n} \right| = \lim_{n\to\infty} \left| \frac{(-1)(2n+1)}{2n+3} \right|$

$= \lim_{n\to\infty} \frac{2n+1}{2n+3} = 1 \Rightarrow R = \frac{1}{L} = 1$

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pt $x=1$: $\sum_{n\geq 0}(-1)^n \cdot \frac{1}{2n+1}$. Notăm: $b_n = \frac{1}{2n+1}$ și folosim criteriul Leibnitz:

$\frac{b_{n+1}}{b_n} = \frac{2n+1}{2n+3} < 1 \Rightarrow b_{n+1} < b_n \to b_n \downarrow_s$ (1)

$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{2n+1} = 0$ (2)

$\xrightarrow{(1)(2)} \sum_{n\geq 0}(-1)^n b_n$ convergentă

pt $x=-1$: $\sum_{n\geq 0}\frac{(-1)^{3n+1}}{2n+1}$ convergentă (se folosește criteriul Leibnitz pt b_n și $\sum_{n\geq 0}(-1)^{3n+1} b_n$)

$\Rightarrow$ M.C.: $[-1,1]$

Pe $[-1,1]$ $\exists$ suma seriei $f:[-1,1]\to\mathbb{R}; f(x) = \sum_{n\geq 0}(-1)^n \frac{x^{2n+1}}{2n+1}$

$f(x) = \sum_{n\geq 0}(-1)^n \cdot \frac{x^{2n+1}}{2n+1}$ | derivez.

$\Rightarrow f'(x) = \sum_{n\geq 0}(-1)^n \cdot x^{2n} = \sum_{n\geq 0}(-x^2)^n = \frac{1}{1+x^2}$ | $\int dx$

$\Rightarrow f(x) = \operatorname{arctg} x + C$

$f(0) = \sum_{n\geq 0}(-1)^n \cdot \frac{0}{2n+1} = 0$

$\Rightarrow f(0) = \underbrace{\operatorname{arctg} 0}_{0} + C = 0 \Rightarrow C = 0$

$\Rightarrow f(x) = \operatorname{arctg} x$

2. $\sum_{n\geq 1}\frac{(-1)^{n-1}}{n(2n-1)} \cdot \left(\frac{1-x}{1-2x}\right)^{2n-1}$. Notăm: $\frac{1-x}{1-2x} = y$

$\Rightarrow \sum_{n\geq 1}\frac{(-1)^{n-1}}{n(2n-1)} y^{2n-1}$. Notăm: $a_n = \frac{(-1)^{n-1}}{n(2n-1)}$

$L = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^{n}(-1)}{(n+1)(2n+1)} \cdot \frac{n(2n-1)}{(-1)^{n-1}}\right| =$

$= \lim_{n\to\infty}\frac{2n^2-n}{2n^2+3n+1} = 1 \Rightarrow R = \frac{1}{L} = 1$

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pt $y=1$: $\sum_{n\geq 1} \frac{(-1)^{n-1}}{n(2n-1)}$. Notăm: $b_n = \frac{1}{n(2n-1)}$ și folosim criteriul Leibnitz.

$$\frac{b_{n+1}}{b_n} = \frac{n(2n-1)}{(n+1)(2n+1)} = \frac{2n^2-n}{2n^2+3n+1} < 1 \Rightarrow b_{n+1} < b_n \Rightarrow b_n \downarrow \text{ (1)}$$

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n(2n-1)} = 0 \text{ (2)}$$

$\xRightarrow{(1)(2)}$ $\sum_{n\geq 1} (-1)^{n-1} b_n$ convergentă

pt $y=-1$: $\sum_{n\geq 1} \frac{(-1)^{n-1}}{n(2n-1)} (-1)^{2n-1} = \sum_{n\geq 1} \frac{(-1)^{3n-2}}{n(2n-1)}$ convergentă

(folosim criteriul lui Leibnitz pt b_n și $\sum_{n\geq 1} (-1)^{3n-2} b_n$)

$\Rightarrow$ M.C.: $[-1,1]$ în y

$$-1 \leq y \leq 1 \Leftrightarrow -1 \leq \frac{1-x}{1-2x} \leq 1$$

$$\text{I}: -1 \leq \frac{1-x}{1-2x} \Leftrightarrow \frac{1-x}{1-2x} + 1 \geq 0 \Leftrightarrow \frac{2-3x}{1-2x} \geq 0 \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} \begin{cases} 2-3x \geq 0 \Rightarrow x \leq \frac{2}{3} \Rightarrow x \in (-\infty, \frac{2}{3}] \\ 1-2x > 0 \Rightarrow x < \frac{1}{2} \Rightarrow x \in (-\infty, \frac{1}{2}) \end{cases} \xRightarrow{\cap} x \in (-\infty, \frac{1}{2}) \\ \begin{cases} 2-3x \leq 0 \Rightarrow x \geq \frac{2}{3} \Rightarrow x \in [\frac{2}{3}, \infty) \\ 1-2x < 0 \Rightarrow x > \frac{1}{2} \Rightarrow x \in (\frac{1}{2}, \infty) \end{cases} \xRightarrow{\cap} x \in [\frac{2}{3}, \infty) \end{cases} \xRightarrow{\cup}$$

$$\Rightarrow x \in (-\infty, \tfrac{1}{2}) \cup [\tfrac{2}{3}, \infty) \text{ (1)}$$

$$\text{II}: \frac{1-x}{1-2x} \leq 1 \Leftrightarrow \frac{1-x}{1-2x} - 1 \leq 0 \Leftrightarrow \frac{x}{1-2x} \leq 0 \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} \begin{cases} x \geq 0 \Rightarrow x \in [0, \infty) \\ 1-2x < 0 \Rightarrow x \in (\frac{1}{2}, \infty) \end{cases} \xRightarrow{\cap} x \in (\frac{1}{2}, \infty) \\ \begin{cases} x \leq 0 \Rightarrow x \in (-\infty, 0] \\ 1-2x > 0 \Rightarrow x \in (-\infty, \frac{1}{2}) \end{cases} \xRightarrow{\cap} x \in (-\infty, 0] \end{cases} \xRightarrow{\cup}$$

$$\Rightarrow x \in (-\infty, 0] \cup (\tfrac{1}{2}, \infty) \text{ (2)}$$

$\xRightarrow[\cap]{(1)(2)}$ M.C: $(-\infty, 0] \cup [\frac{2}{3}, \infty)$ în x.

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Pe $[-1,1]$ $\exists$ suma seriei $f:[-1,1]\to\mathbb{R}$:

$$f(y)=\sum_{n\geq 1}\frac{(-1)^{n-1}}{n(2n-1)}y^{2n-1} \quad | \text{ derivez}$$

$$\Rightarrow f'(y)=\sum_{n\geq 1}\frac{(-1)^{n-1}}{n}\cdot y^{2n-2} \quad |\cdot y^2$$

$$y^2f'(y)=\sum_{n\geq 1}\frac{(-1)^{n-1}}{n}\cdot y^{2n} \quad | \text{ derivez}$$

$$\left[y^2f'(y)\right]'=\sum_{n\geq 1}2\cdot(-1)^{n-1}y^{2n-1}=\sum_{n\geq 1}2y(-1)^{n-1}y^{2(n-1)}=$$

$$=\sum_{n\geq 1}2y(-1)^{n-1}\cdot(y^2)^{n-1}=2y\sum_{n\geq 1}(-y^2)^{n-1}=$$

$$=2y\cdot\frac{1}{1+y^2}=\frac{2y}{1+y^2} \quad | \int dy$$

$$\Rightarrow y^2f'(y)=\int\frac{2y}{1+y^2}dy=\ln(1+y^2)+C \Bigg\} \Rightarrow$$

$$y=0\Rightarrow 0\cdot f'(0)=\ln(1)+C\Rightarrow C=0 \Bigg\}$$

$$\Rightarrow y^2f'(y)=\ln(1+y^2)\Rightarrow f'(y)=\frac{\ln(1+y^2)}{y^2} \quad | \int dy$$

$$\Rightarrow f(y)=\int\frac{\ln(1+y^2)}{y^2}dy=-\frac{1}{y}\ln(1+y^2)+\int\frac{1}{y}\cdot\frac{2y}{1+y^2}dy=$$

$$=-\frac{1}{y}\ln(1+y^2)+2\int\frac{1}{1+y^2}dy=$$

$$=-\frac{1}{y}\ln(1+y^2)+2\operatorname{arctg}y+K \Big\}_{\to K=0} \Rightarrow$$

$$f(0)=\sum_{n\geq 1}\frac{(-1)^{n-1}}{n(2n-1)}\cdot 0=0$$

$$\Rightarrow f(y)=2\operatorname{arctg}y-\frac{1}{y}\ln(1+y^2)\Rightarrow$$

$$\Rightarrow f(x)=2\operatorname{arctg}\left(\frac{1-x}{1-2x}\right)-\frac{1-2x}{1-x}\ln\left(1+\left(\frac{1-x}{1-2x}\right)^2\right)=$$

$$=2\operatorname{arctg}\left(\frac{1-x}{1-2x}\right)-\frac{1-2x}{1-x}\ln\frac{5x^2-6x+2}{(1-2x)^2};$$

$x\in(-\infty,0]\cup\left[\frac{2}{3},\infty\right)$

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3. $\sum_{n\geq 1} \frac{2^n}{2n+1} \cdot x^n$. Notăm: $a_n = \frac{2^n}{2n+1}$

$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{2^{n+1}}{2n+3} \cdot \frac{2n+1}{2^n}\right| =$

$= \lim_{n\to\infty} \frac{4n+2}{2n+3} = 2 \Rightarrow R = \frac{1}{L} = \frac{1}{2}$

pt $x = \frac{1}{2}$: $\sum_{n\geq 1} \frac{2^n}{2n+1} \cdot \left(\frac{1}{2}\right)^n = \sum_{n\geq 1} \frac{1}{2n+1}$. Notăm: $b_n = \frac{1}{2n+1}$

și folosim criteriul comparației la limită.

Fie $c_n = \frac{1}{n^\alpha}$; $\alpha \in \mathbb{R}$

$\Rightarrow \lim_{n\to\infty} \frac{b_n}{c_n} = \lim_{n\to\infty} \frac{n^\alpha}{2n+1} = \lim_{n\to\infty} n^{\alpha-1} \cdot \frac{1}{2+\frac{1}{n}} \neq 0$ și

$\neq \infty \Leftrightarrow \alpha = 1 \Rightarrow$

$\Rightarrow \sum_{n\geq 1} c_n$ divergentă $\Rightarrow \sum_{n\geq 1} b_n$ divergentă.

pt $x = -\frac{1}{2}$: $\sum_{n\geq 1} \frac{2^n}{2n+1} \left(-\frac{1}{2}\right)^n = \sum_{n\geq 1} (-1)^n \cdot \frac{1}{2n+1}$.

Notăm: $u_n = \frac{1}{2n+1}$ și folosim criteriul Leibnitz.

$\frac{u_{n+1}}{u_n} = \frac{2n+1}{2n+3} < 1 \Rightarrow u_{n+1} < u_n \Rightarrow u_n \searrow$ (1)

$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{2n+1} = 0$ (2)

$\xrightarrow{(1)(2)} \sum_{n\geq 1} (-1)^n \cdot u_n$ convergentă

$\Rightarrow$ M.C.: $\left[-\frac{1}{2}, \frac{1}{2}\right)$

Pe $\left[-\frac{1}{2}, \frac{1}{2}\right)$ $\exists$ suma seriei $f(x) = \sum_{n\geq 1} \frac{2^n}{2n+1} \cdot x^n$;

$f: \left[-\frac{1}{2}, \frac{1}{2}\right) \to \mathbb{R}$.

$f(x) = \sum_{n\geq 1} \frac{1}{2n+1} (2x)^n$. Notăm: $2x = t^2$; $t \in (0,1)$ (pt $x \geq 0$)

$\Rightarrow f(t) = \sum_{n\geq 1} \frac{1}{2n+1} \cdot t^{2n} \Big| \cdot t$

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$$t f(t) = \sum_{n\geq 1} \frac{1}{2n+1} t^{2n+1} \;\Big|\; \text{derivez}$$

$$[t f(t)]' = \sum_{n\geq 1} t^{2n} = \sum_{n\geq 1} (t^2)^n = \frac{1}{1-t^2} \;\Big|\; \int dt$$

$$t f(t) = \int \frac{1}{1-t^2} = \frac{-1}{2t} \ln\left|\frac{t-1}{t+1}\right| + C \Rightarrow$$

$$\Rightarrow f(t) = \frac{-1}{2t^2} \ln \frac{1-t}{1+t} \Rightarrow f(x) = \frac{-1}{4x} \ln \frac{1-\sqrt{2x}}{1+\sqrt{2x}}; \; x \in \left[0, \frac{1}{2}\right)$$

pt $x<0$ Notăm: $2x = -t^2$; $t \in (-1,0)$ $\Rightarrow$

$$\Rightarrow f(t) = \sum_{n\geq 1} \frac{(-1)^n}{2n+1} \cdot t^{2n} \;\Big|\; \cdot t$$

$$t f(t) = \sum_{n\geq 1} \frac{(-1)^n}{2n+1} \cdot t^{2n+1} \;\Big|\; \text{derivez}$$

$$[t f(t)]' = \sum_{n\geq 1} (-1)^n t^{2n} = \sum_{n\geq 1} (-1)^n \cdot (t^2)^n = \sum_{n\geq 1} (-t^2)^n =$$

$$= \frac{1}{1+t^2} \;\Big|\; \int dt$$

$$t f(t) = \operatorname{arctg} t + C$$

$$t = 0 \Rightarrow 0 \cdot f(0) = \operatorname{arctg} 0 + C \Rightarrow C = 0 \Big\} \Rightarrow$$

$$\Rightarrow t f(t) = \operatorname{arctg} t \Rightarrow f(t) = \frac{1}{t} \operatorname{arctg} t \Rightarrow$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{-2x}} \operatorname{arctg}(\sqrt{-2x}); \; x \in \left[-\frac{1}{2}, 0\right)$$

4. $\sum_{n\geq 1} \frac{(-1)^n}{5^n \cdot n!} x^{2n-1}$. Notăm: $a_n = \frac{(-1)^n}{5^n \cdot n!}$.

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-1)^{n+1}}{5^{n+1}(n+1)!} \cdot \frac{5^n \cdot n!}{(-1)^n}\right| =$$

$$= \lim_{n\to\infty} \frac{1}{5n+5} = 0 \Rightarrow R = \frac{1}{L} = +\infty \Rightarrow \text{M.C.} : \mathbb{R}.$$

$\Rightarrow$ Pe $\mathbb{R}$ există suma seriei $f: \mathbb{R} \to \mathbb{R}$;

$$f(x) = \sum_{n\geq 1} \frac{(-1)^n}{5^n \cdot n!} x^{2n-1} = \sum_{n\geq 1} \frac{(-1)^n}{n!} \left(\frac{x^2}{5}\right)^n \cdot \frac{1}{x} =$$

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$$= \frac{1}{x} \sum_{n\geq 1} \frac{(-1)^n}{n!} \left(\frac{x^2}{5}\right)^n \quad | \cdot x$$

$$\Rightarrow x f(x) = \sum_{n\geq 1} \frac{(-1)^n}{n!} \left(\frac{x^2}{5}\right)^n$$

Adunăm și scadem primul termen:

$$x f(x) = -1 + \sum_{n\geq 0} \frac{\left(\frac{-x^2}{5}\right)^n}{n!} = -1 + e^{\frac{-x^2}{5}} \quad | : x$$

$$f(x) = \begin{cases} \frac{1}{x}\left(e^{-\frac{x^2}{5}} - 1\right) & ; \text{dacă } x \in \mathbb{R}^* \\ 0 & ; \text{dacă } x = 0 \end{cases}$$

• Studiați convergența seriilor numerice și determinați suma lor, folosind serii de puteri.

1. $\sum_{n\geq 0} \frac{(-1)^n}{3n+1}$. Notăm: $x_n = \frac{1}{3n+1}$ și folosim criteriul Leibnitz:

$$\frac{x_{n+1}}{x_n} = \frac{3n+1}{3n+4} < 1 \Rightarrow x_{n+1} < x_n \Rightarrow x_n \downarrow_0 \ (1)$$

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{3n+1} = 0 \ (2)$$

$\xRightarrow{(1)(2)}$ $\sum_{n\geq 0} (-1)^n x_n$ convergentă. $\Rightarrow$ seria are sumă.

Fie seria $\sum_{n\geq 0} (-1)^n \cdot \frac{1}{3n+1} \cdot x^{3n+1}$. Notăm $a_n = \frac{(-1)^n}{3n+1}$

$$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(-1)^{n+1}}{3n+4} \cdot \frac{3n+1}{(-1)^n}\right| =$$

$$= \lim_{n\to\infty} \frac{3n+1}{3n+4} = 1 \Rightarrow R = \frac{1}{L} = 1$$

pt $x = 1$: $\sum_{n\geq 0} \frac{(-1)^n}{3n+1}$ convergentă (demonstrat anterior)

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pt $x=-1$: $\sum_{n\geq 0}(-1)^n \cdot \frac{1}{3n+1}\cdot(-1)^{3n+1} = \sum_{n\geq 0}\frac{(-1)^{4n+1}}{3n+1}$ convergentă

(folosim criteriul Leibnitz cu x_n și $\sum_{n\geq 0}(-1)^{4n+1}x_n$)

$\Rightarrow$ M.C.: $[-1,1]$

$\Rightarrow$ Pe $[-1,1]$ $\exists$ suma seriei $f:[-1,1]\to\mathbb{R}$:

$$f(x) = \sum_{n\geq 0}(-1)^n\cdot\frac{1}{3n+1}\cdot x^{3n+1} \quad \Big| \text{ derivez}$$

$$f'(x) = \sum_{n\geq 0}(-1)^n\cdot x^{3n} = \sum_{n\geq 0}(-1)^n\cdot(x^3)^n = \sum_{n\geq 0}(-x^3)^n =$$

$$= \frac{1}{1+x^3} \quad \Big| \int dx$$

$$f(x) = \int \frac{1}{1+x^3}dx = \int \frac{1}{(x+1)(x^2-x+1)}dx = \int\left[\frac{A}{x+1}+\frac{Bx+C}{x^2-x+1}\right]dx.$$

Avem: $\frac{1}{(x+1)(x^2-x+1)} = \frac{A}{x+1}+\frac{Bx+C}{x^2-x+1} = \frac{A(x^2-x+1)+(x+1)(Bx+C)}{(x+1)(x^2-x+1)}$

$\Rightarrow 1 = Ax^2-Ax+A+Bx^2+Cx+Bx+C \Rightarrow$

$$\Rightarrow \begin{cases} A+B=0 \Rightarrow B=-A \\ B+C-A=0 \\ A+C=1 \Rightarrow C=1-A \end{cases} \Rightarrow -A+1-A-A=0 \Rightarrow 3A=1 \Rightarrow$$

$$\Rightarrow A=\frac{1}{3} \Rightarrow B=-\frac{1}{3} \Rightarrow$$

$$\Rightarrow C=\frac{2}{3}$$

$$\Rightarrow f(x) = \int\left[\frac{1}{3(x+1)}+\frac{-x+2}{3(x^2-x+1)}\right]dx = \frac{1}{3}\int\frac{1}{x+1}dx - \frac{1}{3}\int\frac{x-2}{x^2-x+1}dx$$

$$\int\frac{1}{1+x}dx = \int[\ln(1+x)]'dx = \ln(x+1)+C$$

$$\int\frac{x-2}{x^2-x+1}dx = \frac{1}{2}\int\frac{2x-4}{x^2-x+1}dx = \frac{1}{2}\int\left[\frac{2x-1}{x^2-x+1}-\frac{3}{x^2-x+1}\right]dx =$$

$$= \frac{1}{2}\int\frac{2x-1}{x^2-x+1}dx - \frac{3}{2}\int\frac{1}{x^2-x+1}dx$$

$$\int\frac{2x-1}{x^2-x+1}dx = \int[\ln(x^2-x+1)]'dx = \ln(x^2-x+1)+C.$$

$$\int\frac{1}{x^2-x+1}dx = \int\frac{1}{x^2-x+\frac{1}{4}+\frac{3}{4}}dx = \int\frac{1}{(x-\frac{1}{2})^2+\frac{3}{4}}dx = \int\frac{1}{t^2+\frac{3}{4}}dt =$$

Notăm: $x-\frac{1}{2}=t \Rightarrow dx=dt$

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$$= \frac{1}{\frac{\sqrt3}{2}} \operatorname{arctg} \frac{t}{\frac{\sqrt3}{2}} + C = \frac{2}{\sqrt3} \operatorname{arctg} \frac{2t}{\sqrt3} + C = \frac{2}{\sqrt3} \operatorname{arctg} \frac{2x-1}{\sqrt3} + C$$

$$\Rightarrow \int \frac{x-2}{x^2-x+1}\,dx = \frac{1}{2}\ln(x^2-x+1) - \frac{3}{2}\cdot\frac{2}{\sqrt3}\operatorname{arctg}\frac{2x-1}{\sqrt3} + C.$$

$$\Rightarrow f(x) = \frac{1}{3}\ln(x+1) - \frac{1}{6}\ln(x^2-x+1) + \frac{\sqrt3}{3}\operatorname{arctg}\frac{2x-1}{\sqrt3} + C =$$

$$= \frac{1}{6}\ln\left[\frac{(x+1)^2}{x^2-x+1}\right] + \frac{1}{\sqrt3}\operatorname{arctg}\frac{2x-1}{\sqrt3} + C.$$

$$f(0) = \sum_{n\geq 0}(-1)^n \cdot \frac{1}{3n+1}\cdot 0 = 0$$

$$f(0) = \underbrace{\frac{1}{6}\ln 1}_{0} + \frac{1}{\sqrt3}\operatorname{arctg}\frac{-1}{\sqrt3} + C = 0 \Rightarrow$$

$$\Rightarrow \frac{1}{\sqrt3}\cdot\frac{-\pi}{6} + C = 0 \Rightarrow C = \frac{\pi}{6\sqrt3}$$

$$\Rightarrow f(x) = \frac{1}{6}\ln\left[\frac{(x+1)^2}{x^2-x+1}\right] + \frac{1}{\sqrt3}\operatorname{arctg}\frac{2x-1}{\sqrt3} + \frac{\pi}{6\sqrt3}.$$

Avem: $\sum_{n\geq 0}\frac{(-1)^n}{3n+1} = f(1) = \frac{1}{6}\ln 2^2 + \frac{1}{\sqrt3}\cdot\operatorname{arctg}\frac{1}{\sqrt3} + \frac{\pi}{6\sqrt3} =$

$$= \frac{1}{3}\ln 2 + \frac{\pi}{6\sqrt3} + \frac{\pi}{6\sqrt3} = \frac{1}{3}\ln 2 + \frac{\pi}{3\sqrt3}.$$

2. $\sum_{n\geq 0}\frac{(n+1)^2}{n!}$. Notăm: $x_n = \frac{(n+1)^2}{n!}$ și folosim criteriul raportului:

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n} = \lim_{n\to\infty}\frac{(n+2)^2}{(n+1)!}\cdot\frac{n!}{(n+1)^2} = \lim_{n\to\infty}\frac{(n+2)^2}{(n+1)^3} = 0 < 1$$

$\Rightarrow \sum_{n\geq 0} x_n$ convergentă $\Rightarrow$ seria are sumă.

Fie $e^x = \sum_{n\geq 0}\frac{x^n}{n!} \mid \cdot x$

$\Rightarrow x e^x = \sum_{n\geq 0}\frac{x^{n+1}}{n!} \mid$ derivez

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$$e^x + xe^x = (x+1)e^x = \sum_{n\geq 0} \frac{(n+1)x^n}{n!} \quad | \cdot x$$

$$e^x(x^2+x) = \sum_{n\geq 0} \frac{(n+1)x^{n+1}}{n!} \quad | \text{ derivez}$$

$$e^x(x^2+x) + e^x(2x+1) = e^x(x^2+3x+1) = \sum_{n\geq 0} \frac{(n+1)^2}{n!} \cdot x^n$$

Pentru $x=1 \Rightarrow \sum_{n\geq 0} \frac{(n+1)^2}{n!} = e^1(1+3+1) = 5e$

3. $\sum_{n\geq 1} \frac{(-1)^n}{n}\left(\frac{1}{2}\right)^{2n-1}$. Notăm: $x_n = \frac{(-1)^n}{n}\left(\frac{1}{2}\right)^{2n-1}$ și

folosim criteriul Leibnitz.

$x_n = \frac{(-1)^n}{n} \cdot \left(\frac{1}{2}\right)^{2n-1}$. Notăm: $u_n = \frac{\left(\frac{1}{2}\right)^{2n-1}}{n}$

$$\frac{u_{n+1}}{u_n} = \frac{\left(\frac{1}{2}\right)^{2n+1} \cdot n}{n+1 \cdot \left(\frac{1}{2}\right)^{2n-1}} = \frac{\frac{1}{4}n}{n+1} < 1 \Rightarrow u_{n+1} < u_n \Rightarrow u_n \downarrow s \ (1)$$

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{\left(\frac{1}{2}\right)^{2n-1}}{n} = 0 \ (2)$$

$\overset{(1)(2)}{\Longrightarrow} \sum_{n\geq 1} x_n$ convergentă $\Rightarrow$ seria are sumă.

Fie: $\sum_{n\geq 1} \frac{(-1)^n}{n} \cdot x^{2n-1}$. Notăm: $a_n = \frac{(-1)^n}{n}$

$$L = \left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^{n+1}}{n+1} \cdot \frac{n}{(-1)^n}\right| = \lim_{n\to\infty} \frac{n}{n+1} = 1 \Rightarrow R = \frac{1}{L} = 1$$

pt $x=1 \Rightarrow \sum_{n\geq 1} \frac{(-1)^n}{n}$. Notăm: $b_n = \frac{1}{n}$ și aplicăm

criteriul lui Leibnitz:

$$\frac{b_{n+1}}{b_n} = \frac{n}{n+1} < 1 \Rightarrow b_{n+1} < b_n \Rightarrow b_n \downarrow s \ (1)$$

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n} = 0 \ (2)$$

$\overset{(1)(2)}{\Longrightarrow} \sum_{n\geq 1} (-1)^n \cdot b_n$ convergentă

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pt $x=-1$: $\sum_{n\geq 1} \frac{(-1)^{2n}}{n} = \sum_{n\geq 1} \frac{1}{n}$ divergentă (serie armo. generalizată cu $\alpha=1$)

$\Rightarrow$ M. C. : $(-1,1]$

Pe $(-1,1]$ $\exists$ suma seriei $f:(-1,1]\to\mathbb{R}$;

$$f(x) = \sum_{n\geq 1} \frac{(-1)^n}{n} \cdot x^{2n-1} = \frac{1}{x} \sum_{n\geq 1} \frac{(-1)^n}{n} (x^2)^n \quad \Big| \cdot (-1)$$

$$-f(x) = \frac{1}{x} \sum_{n\geq 1} \frac{(-1)^{n+1}}{n} \cdot (x^2)^n = \frac{1}{x} \cdot \ln(1+x^2) \Rightarrow$$

$$\Rightarrow f(x) = \begin{cases} -\frac{\ln(1+x)}{x} & ; x\in(-1,1]\setminus\{0\} \\ 0 & ; x=0 \end{cases}$$

pt $x=\frac{1}{2}$ $\Rightarrow -2\sum_{n\geq 1} \frac{(-1)^{n+1}}{n} \left(\frac{1}{2}\right)^{2n} = \sum_{n\geq 1} \frac{(-1)^n}{n} \left(\frac{1}{2}\right)^{2n-1} =$

$$= 2\ln\left(\frac{4}{5}\right)$$

4. $\sum_{n\geq 0} \frac{3n^3-n^2+1}{n!}$. Notăm : $x_n = \frac{3n^3-n^2+1}{n!}$ și

Folosim criteriul raportului :

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to\infty} \frac{3(n+1)^3-(n+1)^2+1}{\cancel{(n+1)!}\cdot n+1} \cdot \frac{\cancel{n!}}{3n^3-n^2+1} =$$

$$= \lim_{n\to\infty} \frac{3(n+1)^2 - n - 1 + \frac{1}{n+1}}{3n^3-n^2+1} = 0 < 1 \Rightarrow$$

$\Rightarrow \sum_{n\geq 0} x_n$ convergentă $\Rightarrow$ seria are sumă.

$$\sum_{n\geq 0} \frac{3n^3-n^2+1}{n!} = 3\underbrace{\sum_{n\geq 0} \frac{n^3}{n!}}_{S_1} - \underbrace{\sum_{n\geq 0} \frac{n^2}{n!}}_{S_2} + \underbrace{\sum_{n\geq 0} \frac{1}{n!}}_{S_3}$$

$$e^x = \sum_{n\geq 0} \frac{x^n}{n!} \quad \Big| \text{ derivăm}$$

$$e^x = \sum_{n\geq 0} \frac{nx^{n-1}}{n!} \quad \Big| \cdot x$$

$$xe^x = \sum_{n\geq 0} \frac{nx^n}{n!} \quad \Big| \text{ derivăm}$$

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$(x+1)e^x = \sum_{n\geq 0} \frac{n^2 x^{n-1}}{n!} \mid \cdot x$

$(x^2+x)e^x = \sum_{n\geq 0} \frac{n^2 x^n}{n!} \mid$ derivăm

$(2x+1)e^x + (x^2+x)e^x = (x^2+3x+1)e^x = \sum_{n\geq 0} \frac{n^3 x^{n-1}}{n!}$

pentru $x=1 \Rightarrow 5e = \sum_{n\geq 0} \frac{n^3}{n!} = S_1$; $2e = \sum_{n\geq 0} \frac{n^2}{n} = S_2$

$e^x = \sum_{n\geq 0} \frac{x^n}{n!} \Rightarrow$ pt $x=1 \Rightarrow e = \sum_{n\geq 0} \frac{1}{n!} = S_3$

$\Rightarrow \sum_{n\geq 0} \frac{3n^3-n^2+1}{n!} = 3\cdot 5e - 2e + e = 14e.$

• Dezvoltați în serie de puteri următoarele funcții, pe mulțimea dată.

1. $f(x) = \ln\sqrt{\frac{1+x}{1-x}}$; $x \in (-1,1)$

$f(x) = \ln\sqrt{\frac{1+x}{1-x}} = \ln\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}} = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right) = \frac{1}{2}\sum_{n\geq 0} \frac{x^{2n+1}}{2n+1}.$

2. $f(x) = \frac{3x}{x^2+5x+6}$; $x \in \mathbb{R}\setminus\{-2,-3\}$

$f(x) = \frac{3x}{x^2+5x+6} = \frac{3x}{(x+2)(x+3)} = \frac{A}{x+2} + \frac{B}{x+3}$

$\Rightarrow 3x = A(x+3) + B(x+2) = Ax + 3A + Bx + 2B \Leftrightarrow$

$\Leftrightarrow \begin{cases} A+B=3 \mid \cdot 3 \\ 3A+2B=0 \end{cases} \rightarrow \begin{cases} 3A+3B=9 \\ 3A+2B=0 \end{cases} \Rightarrow B=9 \rightarrow A=-6$

$\Rightarrow f(x) = \frac{-6}{x+2} + \frac{9}{x+3} = -3\cdot\frac{2}{x+2} + 3\cdot\frac{3}{x+3} =$

$= -3\cdot\left(\frac{1}{1+\frac{x}{2}}\right) + 3\left(\frac{1}{1+\frac{x}{3}}\right) = -3\left(1+\frac{x}{2}\right)^{-1} + 3\left(1+\frac{x}{3}\right)^{-1}$

$-3\left(1+\frac{x}{2}\right)^{-1} = -3\sum_{n\geq 0} (-1)^n \cdot \frac{x^n}{2^n}$; $|x|<2$

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$$3\left(1+\frac{x}{3}\right)^{-1} = 3\sum_{n\geq 0}(-1)^n\cdot\frac{x^n}{3^n};\ |x|<3$$

$$\Rightarrow f(x) = 3\sum_{n\geq 0}(-1)^n\cdot\frac{x^n}{3^n} - 3\sum_{n\geq 0}(-1)^n\cdot\frac{x^n}{2^n} =$$

$$= 3\sum_{n\geq 0}(-1)^n\left(\frac{1}{3^n}-\frac{1}{2^n}\right)x^n;\ |x|<2$$

2. $f(x) = \int_0^x \frac{\operatorname{arctg} t}{t}\,dt;\ x\in[-1,1]$

$$(\operatorname{arctg} t)' = \frac{1}{1+t^2} = \sum_{n\geq 0}(-t^2)^n = \sum_{n\geq 0}(-1)^n t^{2n} \Big|\ \int dt$$

$$\operatorname{arctg} t = \sum_{n\geq 0}(-1)^n\frac{t^{2n+1}}{2n+1} + C$$

$$t=0 \Rightarrow \operatorname{arctg} 0 = 0 = \sum_{n\geq 0}(-1)^n\cdot 0 + C \Rightarrow C=0$$

$$\Rightarrow \operatorname{arctg} t = \sum_{n\geq 0}(-1)^n\cdot\frac{t^{2n+1}}{2n+1}\ \Big|\cdot\frac{1}{t}$$

$$\frac{\operatorname{arctg} t}{t} = \sum_{n\geq 0}(-1)^n\cdot\frac{t^{2n}}{2n+1}\ \Big|\ \int_0^x dt;\ x\in[-1,1]$$

$$\int_0^x \frac{\operatorname{arctg} t}{t}dt = \sum_{n\geq 0}(-1)^n\frac{1}{2n+1}\int_0^x t^{2n}dt =$$

$$= \sum_{n\geq 0}(-1)^n\cdot\frac{1}{2n+1}\cdot\frac{t^{2n+1}}{2n+1}\Big|_0^x =$$

$$= \sum_{n\geq 0}(-1)^n\cdot\frac{1}{2n+1}\left(\frac{x^{2n+1}}{2n+1}-0\right) =$$

$$= \sum_{n\geq 0}\frac{(-1)^n}{(2n+1)^2}\cdot x^{2n+1};\ x\in[-1,1]$$

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4. $f(x) = \frac{1}{2x-3}$; $x \in \mathbb{R} \setminus \{\frac{3}{2}\}$

$f(x) = \frac{1}{2x-3} = \frac{-1}{3-2x} = \frac{-1}{1+2(1-x)} = -\left[1+2(1-x)\right]^{-1} =$

$= -\sum_{n\geq 1} (-1)^n \left[2(1-x)\right]^n = -\sum_{n\geq 1} 2^n (x-1)^n$; $|x-1|<1 \Leftrightarrow$

$\Leftrightarrow x \in (0,2)$

- Calculați $\int_0^1 e^{-x^2} dx$ cu 4 zecimale exacte.

Avem $e^y = \sum_{n\geq 0} \frac{y^n}{n!}$; $y \in \mathbb{R}$. Alegem $y = -x^2 \Rightarrow$

$\Rightarrow e^{-x^2} = \sum_{n\geq 0} \frac{(-x^2)^n}{n!} = \sum_{n\geq 0} \frac{(-1)^n}{n!} \cdot x^{2n} \Big| \int_0^1 dx$

$\int_0^1 e^{-x^2} dx = \sum_{n\geq 0} \frac{(-1)^n}{n!} \cdot \frac{x^{2n+1}}{2n+1} \Big|_0^1 = \sum_{n\geq 0} (-1)^n \cdot \frac{1}{n!(2n+1)}$

Notăm: $x_n = \frac{1}{n!(2n+1)}$ și folosim criteriul Leibnitz:

$\frac{x_{n+1}}{x_n} = \frac{n!(2n+1)}{(n+1)!(2n+3)} = \frac{2n+1}{(n+1)(2n+3)} < 1 \Leftrightarrow x_{n+1} < x_n \Rightarrow$

$\Rightarrow x_n \searrow$ (1)

$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{n!(2n+1)} = 0$ (2)

(1)(2) $\Rightarrow \sum_{n\geq 0} (-1)^n x_n$ convergentă $\Rightarrow$ seria are sumă.

Notăm S = suma seriei și S_n = șirul sertilor parțiale.

Trebuie să det cel mai mic $n \in \mathbb{N}$ a.î. $|S - S_n| < x_{n+1} <$

$< 10^{-5}$.

$x_1 = \frac{1}{1!\cdot 3} = \frac{1}{3}$; $x_2 = \frac{1}{2!\cdot 5} = \frac{1}{10}$; $x_3 = \frac{1}{3!\cdot 7} = \frac{1}{42}$; $x_4 = \frac{1}{4!\cdot 9} =$

$= \frac{1}{216}$; $x_5 = \frac{1}{5!\cdot 11} = \frac{1}{1320}$; $x_6 = \frac{1}{6!\cdot 13} = \frac{1}{9360}$;

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$$x_7 = \frac{1}{7!\cdot 15} = \frac{1}{75600} \; ; \; x_8 = \frac{1}{8!\cdot 17} = \frac{1}{685400} < \frac{1}{10^5}$$

$\Rightarrow m+1 = 8 \Rightarrow m = 7$

$\Rightarrow |S - S_7| < x_8 \Rightarrow -x_8 < S - S_7 < x_8 \; | + S_7$

$S_7 - x_8 < S < S_7 + x_8$

$S_7 = x_0 - x_1 + x_2 - x_3 + x_4 - x_5 + x_6 - x_7 =$

$= 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \frac{1}{1320} + \frac{1}{9360} - \frac{1}{75600} =$

$\cong 0,746823$

$x_8 = \frac{1}{685400} \cong 0,000001459$

$\Rightarrow 0,746823 - 0,000001459 < S < 0,746823 + 0,000001459$

$\Rightarrow 0,746822 < S < 0,746824$

$\Rightarrow \int_0^1 e^{-x^2} dx = 0,7468\ldots$

- Calculați $\int_0^1 \sin(x^3)\,dx$ cu 3 zecimale exacte.

Avem $\sin x = \sum_{n\geq 0} (-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!} \; ; \; x \in \mathbb{R}$

$x \to x^3 \Rightarrow \sin(x^3) = \sum_{n\geq 0} (-1)^n \cdot \frac{x^{6n+3}}{(2n+1)!} \Big| \int_0^1 dx$

$\Rightarrow \int_0^1 \sin(x^3)\,dx = \sum_{n\geq 0} (-1)^n \cdot \frac{1}{(2n+1)!} \cdot \frac{x^{6n+4}}{6n+4} \Big|_0^1 =$

$= \sum_{n\geq 0} (-1)^n \cdot \frac{1}{(2n+1)!(6n+4)}$

Notăm: $x_n = \frac{1}{(2n+1)!(6n+4)}$ și folosim criteriul Leibnitz.

$\frac{x_{n+1}}{x_n} = \frac{(2n+1)!(6n+4)}{(2n+3)!(6n+10)} < 1 \Rightarrow x_{n+1} < x_n \Rightarrow x_n \downarrow_S (1)$

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$\lim_{n\to\infty} x_n = \lim_{n\to\infty} \frac{1}{(2n+1)!(6n+4)} = 0$ (2)

(1)(2) $\Rightarrow \sum_{n\geq 0} (-1)^n x_n$ convergentă $\Rightarrow$ seria are sumă

Notăm: $a_n = (-1)^n x_n$.

S = suma seriei ; S_n = șirul sumelor parțiale.

Trebuie găsit cel mai mic $n \in \mathbb{N}$ a.î. $|S - S_n| < a_{n+1} < 10^{-4}$

$a_1 = \frac{1}{3! \cdot 10} = \frac{1}{60}$; $a_0 = \frac{1}{1! \cdot 4} = \frac{1}{4}$;

$a_2 = \frac{1}{5! \cdot 16} = \frac{1}{1920}$; $a_3 = \frac{1}{7! \cdot 22} = \frac{1}{110880} < \frac{1}{10^4}$

$\Rightarrow n+1 = 3 \Rightarrow n = 2$

$\Rightarrow |S - S_2| < a_3 \Rightarrow -a_3 < S - S_2 < a_3 \mid + S_2$

$S_2 - a_3 < S < S_2 + a_3$

$S_2 = a_0 - a_1 + a_2 = \frac{1}{4} - \frac{1}{60} + \frac{1}{1920} \cong 0{,}23385$

$a_3 \cong 0{,}00001$

$\Rightarrow 0{,}23385 - 0{,}00001 < S < 0{,}23385 + 0{,}00001$

$0{,}23384 < S < 0{,}23386$

$\Rightarrow \int_0^1 \sin(x^3)\,dx = 0{,}2338\ldots$

- Calculați $\int_0^{1/2} \frac{1}{\sqrt{1+x^4}}\,dx$ cu 4 zecimale exacte

Folosim dezvoltarea binomială cu $y = x^4$ și $\alpha = -\frac{1}{2}$

$$\frac{1}{\sqrt{1+x^4}} = 1 + \sum_{n\geq 1} \frac{-\frac{1}{2}\cdot\left(-\frac{3}{2}\right)\cdot\left(-\frac{5}{2}\right)\cdots\left(-\frac{1}{2}-(n-1)\right)}{n!}\cdot (x^4)^n =$$

$$= 1 + \sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot\ldots\cdot(2n-1)}{2^n \cdot n!} \cdot x^{4n}$$

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$L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$, unde $a_n \overset{not}{=} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2^n \cdot n!}$

$L = \lim_{n\to\infty} \left| \frac{(-1)^{n+1+(-1)}}{(-1)^n} \cdot \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)(2n+1)}{2^{n+1}\cdot (n+1)!} \cdot \frac{2^n \cdot n!}{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)} \right| =$

$= \lim_{n\to\infty} \frac{2n+1}{2n+2} = 1 \Rightarrow R = \frac{1}{L} = 1$

pt $x=1$: $\sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2^n \cdot n!}$. Notăm: $u_n = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2^n \cdot n!}$

și folosim criteriul Leibnitz:

$\frac{u_{n+1}}{u_n} = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)(2n+1)}{2\cdot 2^{n+1}\cdot (n+1)!} \cdot \frac{2^n \cdot n!}{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)} =$

$= \frac{2n+1}{2n+2} < 1 \Rightarrow u_{n+1} < u_n \Rightarrow u_n \downarrow_s (a)$

$\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2^n \cdot n!}$; $0 < u_n$; $\forall n \geq 1$

Pentru $\forall k \in \mathbb{N} \Rightarrow k^2 + 2k < k^2 + 2k + 1 \Leftrightarrow k(k+2) < (k+1)^2 \Rightarrow$

$\Rightarrow \frac{k}{k+1} < \frac{k+1}{k+2}$; $\forall k \in \mathbb{N}$

$\Rightarrow u_n = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2^n \cdot n!} = \frac{1\cdot 3\cdot 5\cdot \ldots \cdot (2n-1)}{2\cdot 4\cdot 6\cdot \ldots \cdot (2n)} =$

$= \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot \ldots \cdot \frac{2n-1}{2n} < \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdot \ldots \cdot \frac{2n}{2n+1} \quad \Big| \cdot u_n$

$u_n^2 < \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot \ldots \cdot \frac{2n-1}{2n} \cdot \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdot \ldots \cdot \frac{2n}{2n+1} = \frac{1}{2n+1} \quad \Big| \sqrt{}$

$\Rightarrow u_n < \frac{1}{\sqrt{2n+1}} \Rightarrow 0 < u_n < \frac{1}{\sqrt{2n+1}}$ (1)

$\lim_{n\to\infty} 0 = 0$ (2)

$\lim_{n\to\infty} \frac{1}{\sqrt{2n+1}} = 0$ (3)

$\xrightarrow[\text{crit. cleşte.}]{(1)(2)(3)} \lim_{n\to\infty} u_n = 0$ (b)

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(a)(b) $\Rightarrow \sum_{n\geq 1} a_n$ convergentă

pt $x=-1$: $\sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^n \cdot n!} = \sum_{n\geq 1} a_n$ convergentă

$\Rightarrow$ M. C. : $[-1,1]$

$x\in[-1,1] \Rightarrow x^4 \in [0,1]$

$$\Rightarrow \frac{1}{\sqrt{1+x^4}} = 1 + \sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^n \cdot n!} \cdot x^{4n}$$

$$\int_0^{1/2} \frac{1}{\sqrt{1+x^4}}\,dx = \frac{1}{2} + \sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^n \cdot n!} \cdot \frac{1}{2^{4n+1}(4n+1)} =$$

$$= \frac{1}{2} + \sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^{5n+1} \cdot n!} \cdot \frac{1}{4n+1}$$

Notăm: $b_n = \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^{5n+1}\cdot n!} \cdot \frac{1}{4n+1}$ și folosim criteriul Leibnitz :

$$\frac{b_{n+1}}{b_n} = \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)(2n+1)}{2^5 \cdot 2^{5n+5}\,(n+1)!\,(4n+5)} \cdot \frac{2^{5n+1}\cdot n!\cdot (4n+1)}{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)} =$$

$$= \frac{(2n+1)(4n+1)}{2^5\cdot (n+1)(4n+5)} = \frac{8n^2+6n+1}{128n^2+288n+160} < 1 \Rightarrow$$

$\Rightarrow b_{n+1} < b_n \Rightarrow b_n \searrow$ (1)

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^{5n+1}\cdot n!} \cdot \frac{1}{4n+1} =$$

$$= \lim_{n\to\infty} \underbrace{\frac{1\cdot 3\cdot 5\cdot \ldots\cdot (2n-1)}{2^n \cdot n!}}_{\to 0} \cdot \underbrace{\frac{1}{(4n+1)2^{4n+1}}}_{\to 0} = 0 \quad (2)$$

(1)(2) $\Rightarrow \sum_{n\geq 1} (-1)^n \cdot b_n$ convergentă $\Rightarrow$ seria are sumă.

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Notăm: S_n = șirul sumelor parțiale și S = suma seriei
Trebuie găsit cel mai mic $n \in \mathbb{N}$ a.î. $|S - S_n| < a_{n+1} < 10^{-5}$.

$a_0 = \frac{1}{2}$; $a_1 = + \frac{1}{320}$; $a_2 = \frac{1}{12288}$; $a_3 = \frac{15}{13 \cdot 2^{17}} < \frac{1}{10^5}$

$\Rightarrow n+1 = 3 \Rightarrow n = 2$

$\Rightarrow |S - S_2| < a_3 \Rightarrow -a_3 < S - S_2 < a_3 \mid + S_2$

$S_2 - a_3 < S < S_2 + a_3$

$S_2 = a_0 - a_1 + a_2 = \frac{1}{2} - \frac{1}{320} + \frac{1}{12288} \cong 0,4969$

$\Rightarrow a_3 = \frac{5}{13 \cdot 2^{17}} \cong 0,000003$

$\Rightarrow 0,4969\ldots - 0,000003 < S < 0,4969\ldots + 0,000003$

$0,4969\ldots < S < 0,4969\ldots$

$\Rightarrow S \cong 0,4969$ (cu 4 zecimale)

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